

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Developed the search back end of a debate research web application in *Python (FastAPI)* that indexes **600K+ debate-evidence documents** on *Elasticsearch*, returning results in **under 1 second** for **10K+ site visitors**
- Optimized the application by indexing **600K+ documents** on *Elasticsearch*, returning results in **under 1 second**
- Enhanced the application with recency-based relevance so the current month's debate topic ranks first in search
- Followed the SDLC in an *Agile* team of **10 engineers**, from design and development to deployment on *AWS*

Product Development Intern | Jetson

January 2025 – February 2025

- Documented UI/UX test findings in a **5-page report** on strategies to optimize user engagement for the CEO
- Gathered feedback across **7 iterations of UI/UX testing** on **3 experimental features** for **100K+ users**
- Cut the application's onboarding from **12 to 5 pages** in *Figma* and validated the redesign through **A/B testing**

Data Analytics Intern | STEM E

June 2024 – July 2024

- Processed **200+ TikTok posts** spanning **3 years** in *Python (pandas)*, then charted trends in *Power BI* and *Excel*
- Found through data analysis that "science fun facts" posts earned **69.3%** of views from just **7%** of all content
- Communicated the data insight as a "fun facts" strategy the Executive Director adopted, lifting engagement **24%**

PROJECTS

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Configured a relational database (*PostgreSQL* on *Supabase*) and wrote *SQL* queries by field, year and citation count
- Developed a web application that graphs how a paper's outbound citations reach **26 research fields** (*React Flow*)
- Integrated the *OpenAlex* and *Semantic Scholar* APIs to resolve each paper and backfill its missing metadata
- Added a **RAG** agent (*Claude API*) to the application that answers from session-scoped vector documents

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built a client server system: a *Python FastAPI* back end ingesting water, waste and energy data from all **3 tools**
- Developed a *Next.js* dashboard that helps schools monitor and manage the data flowing through the system
- Analyzed data with **4 Random Forest models** to flag anomalies and give actionable steps via a *Gemma* Voice LLM
- Directed a team of 5 from ideation and development to the final presentation. **USAII Global Hackathon Finalist**

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Built a web application that auto-books meetings for **30+ people**. **Won 2025 Congressional App Challenge**
- Tested the *React/Firebase* application across **14 unit and integration test cases**, from availability scan to booking
- Solved the scheduling problem with a **5-step Python** algorithm: scan calendars, rank slots, email confirmations
- Developed a lobby feature in the application that authenticates invited participants and queues follow-up meetings

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Analyzed **15,000** truck-listing images and cleaned the data by de-duplicating listings and filtering non-USD prices
- Tested and optimized the extraction, fixing each wrong prediction until **10 consecutive images** were right
- Developed the extraction application: a *JSON schema* guiding a *Claude Sonnet 4.6* agent to read truck photos
- Built analytical *quantile regression* model on extracted data that priced trucks with **87% accuracy** on **30 samples**

SKILLS

Programming Languages: Java, SQL, Python, C++, JavaScript, HTML/CSS

Frameworks/Platforms: PostgreSQL, MySQL, Supabase, FastAPI, React, Next.js, AWS, Node.js, Flask, Firebase

DevTools: Git, GitHub, CI/CD, Docker, Catch2, Clang, VS Code, Claude Code, Codex, Figma, Bun, Arduino IDE

Data & Analytics: Microsoft Excel, Power Point, pandas, NumPy, Power BI, A/B Testing, Hootsuite

Concepts/Practices: Application Development, Databases, Client-Server Architecture, SDLC, Agile, Testing, OOP